ASSESSOR'S MANUAL

VOLUME II COMMERCIAL







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STATE OF MICHIGAN
DEPARTMENT OF TREASURY
LANSING

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GOVERNOR

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STATE TREASURER

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SELECTION OF METHOD

METHOD AVAILABLE – The Michigan Assessor's Manual offers the user the choice of three methods of computing reproduction or replacement costs of buildings: the Calculator Method, Segregated Cost Method and the Comparative Cost Indexes and Multipliers. In addition there are many Unit-in-Place costs, with which the user may build up or adjust costs or price miscellaneous construction items.

1. **The Calculator Method –** These pages contain average square foot costs for various classes, occupancy types and qualities of buildings, together with modifiers for common deviations from the descriptions of the typical buildings listed. Instructions for the use of the Calculator Method and step-by-step examples are given in "Introduction to the Calculator Method". Modifiers to use with the method are found in each occupancy section. The listed costs are averages including architects' and engineers' fees, contractors' profit and overhead, permits, sales tax, insurance during construction, interest on construction financing and miscellaneous costs.

2. **The Segregated Cost Method –** Sections SEG 1 through SEG 8. These sections give the cost per square foot of the major building components (foundation, frame, floor, walls, etc.). Instructions for the use of the Segregated Cost Method are contained in "Introduction to the Segregated Cost Method", and Unit-in-Place costs are given in Sections UIP 1 through UIP 8 for adjusting or building up component costs and for pric- ing miscellaneous items. Additional supplemental costs are included in Sections UIP 11 through UIP 18. These costs of component items include labor, material, and a pro rata share of the additional costs of construction, except for architects' fees which are generally excluded, except where noted. If it is desired to include an architect's fee, schedules of typical fees are listed in Appendix C.

3. **Comparative Cost Indexes and Multipliers –** Appendix B. These tables give indexes and multipliers by which known historical costs may be converted directly to present-day costs.

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CLASS OF CONSTRUCTION

The Class of Construction is the basic subdivision in the Michigan Assessor's Manual, dividing all buildings into five basic cost groups and two subgroups by type of framing (supporting co umns and beams), walls, floors and roof structures, and fireproofing.

Class A buildings have fireproofed structural steel frames with reinforced concrete or masonry floors and roofs.

Class B buildings have reinforced concrete frames and concrete or masonry floors and roofs.

Class C buildings have masonry or concrete exterior walls, and wood or steel roof and floor structures, except for concrete slab on grade.

Class D buildings generally have wood frame, floor, and roof structure. They may have a concrete floor on grade and other substitute materials, but are considered combustible construction. This class includes the pre-engineered pole- or post-frame buildings.

Class S buildings have frames, roofs and walls of incombustible metal. This class includes all the pre-engineered metal buildings.

In each class there will be variations, combinations and subclasses, but for purposes of pricing, the major elements of the building should be considered in selecting costs from the tables. Thus, if a building, which is otherwise in Class B, has a wood or steel truss roof, the costs for the Class B building may still be representative, or a Class C building may have concrete plank floors. Interpolations may be made if the assessor feels the building overlaps two classes sufficiently or the Segregated Cost Sections may be used to modify the cost.

In most localities, some buildings are built which are hybrids in construction, such as those with complete Class A framing, including columns and girders, but with wood floor joists and sheathing. In these cases, the assessor must judge whether to adjust the costs or interpolate between classes and qualities.

Further details and sketches of the various construction types will be found on Pages 3 through 7 of this section, as well as in Section UIP 1, which has definitions and sketches of framing types.

CLASS OF CONSTRUCTION INDICATORS

CLASS	FRAME	FLOOR	ROOF	WALLS
A	Structural steel columns and beams, fireproofed with masonry, concrete, plaster, or other non-combustible material.	Concrete or concrete on steel deck, fireproofed.	Formed concrete, precast slabs, concrete or gypsum on steel deck, fireproofed.	Nonbearing curtain walls, masonry, concrete, metal and glass panels, stone, steel studs and masonry, tile or stucco, etc.
B	Reinforced concrete columns and beams. Fire-resistant construction.	Concrete or concrete on steel deck, fireproofed.	Formed concrete, precast slabs, concrete or gypsum on steel deck, fireproofed.	Nonbearing curtain walls, masonry, concrete, metal and glass panels, stone, steel studs and masonry, tile or stucco, etc.
C	Masonry or concrete load-bearing walls with or without pilasters. Masonry, concrete or curtain walls with full or partial open steel, wood, or concrete frame.	Wood or concrete plank on wood or steel floor joists, or concrete slab on grade.	Wood or steel joists with wood or steel deck. Concrete plank.	Brick, concrete block, or tile masonry, tilt-up, formed con- crete, nonbearing curtain walls.
CMILL	Masonry load-bearing walls with or without pilasters. Masonry walls with heavy wood timber frame."	Mill or laminated wood floor or concrete slab on grade.	Wood joists and trusses with wood deck.	Heavy brick, block or tile masonry.
D	Wood or steel studs in bearing wall, full or partial open wood or steel frame, primarily combustible construction.	Wood or steel floor joists or concrete slab on grade.	Wood or steel joists with wood or steel deck.	Almost any material except bearing or curtain walls of solid masonry or concrete. Generally combustible con- struction.
DPOLE	Wood posts or poles and trussed rafters.	Wood joists and deck or concrete slab on grade.	Metal skin on wood purlins or nailers.	Metal skin on wood girts or nailers.
S	Metal bents, columns, girders, purlins and girts without fire- proofing, incombustible construction.	Wood or steel deck on steel floor joists, or concrete slab on grade.	Steel or wood deck on steel joists.	Metal skin or sandwich panels. Generally incombustible.

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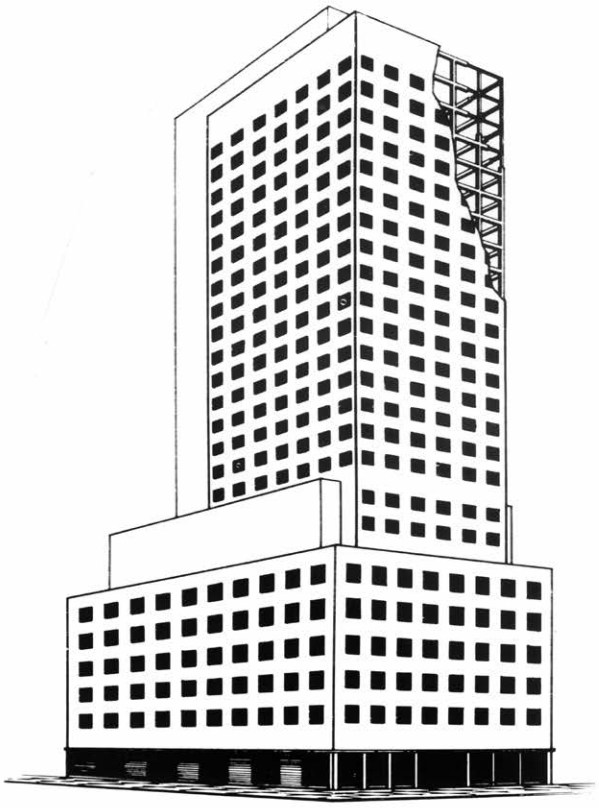
CLASS A BUILDINGS

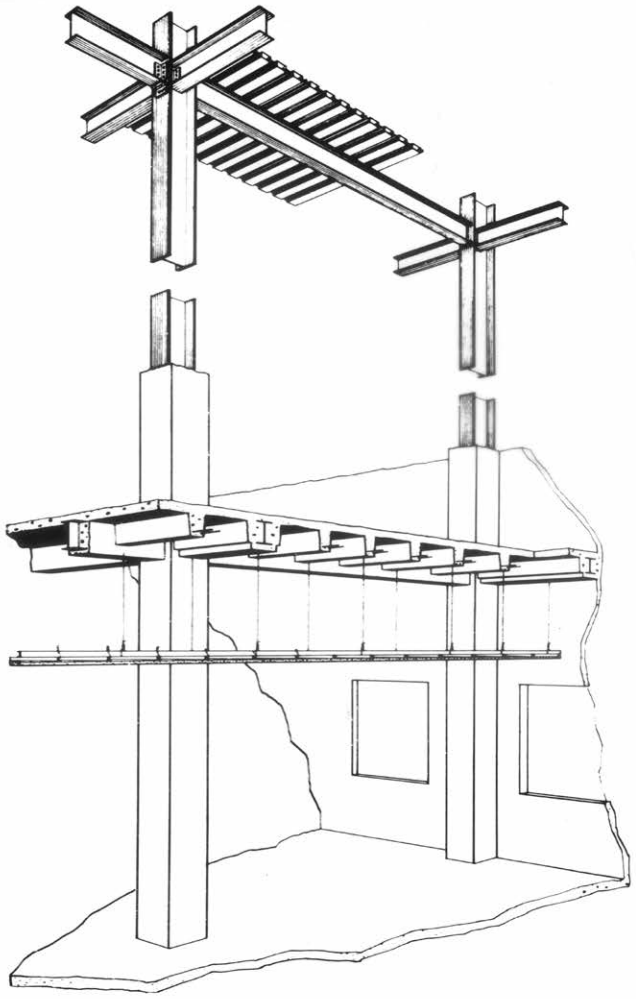
The primary feature of Class A buildings is the fireproofed, protected structural steel frame, which may be welded, bolted, or riveted together. The fireproofing may be masonry, poured concrete, plaster, sprayed fiber, or any other type which will give a high fire-resistance rating.

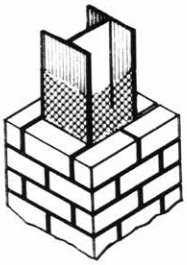
Floors and roofs in Class A structures are normally reinforced concrete on steel decking or formed slabs resting on the frame or poured so as to become integral with it. They may also be composed of prefabricated panels and may be mechanically stressed.

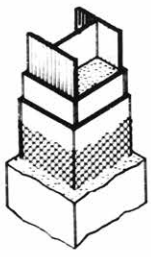
Exterior walls will be curtain walls of masonry, concrete, steel studs and masonry, tile or stucco, or one of the many types of panels of metal, glass, concrete, and other materials. Interior partitions will frequently be of masonry or gypsum block although many movable and lightweight partitions are used.

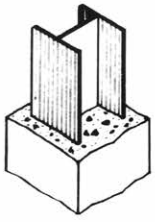
Included in this classification are Uniform, Basic and Standard Building Code construction, Types I and II (noncombustible). This class is also referred to as Modified Fire Resistive or Two – Four Hour Construction.











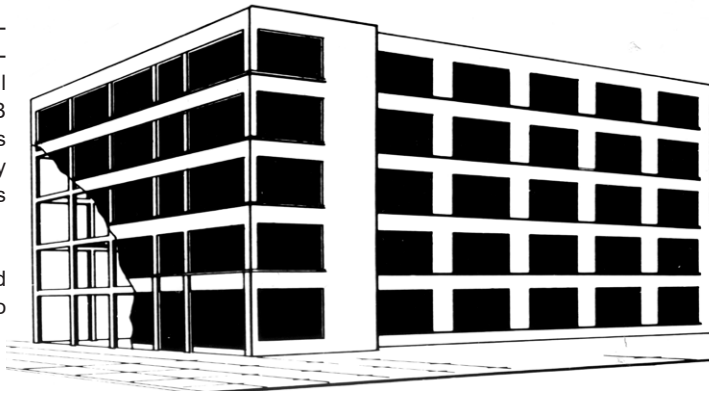
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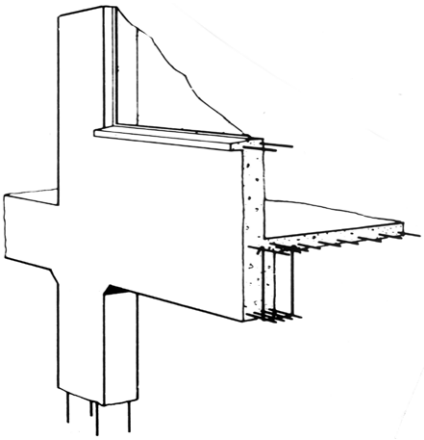
CLASS B BUILDINGS

The primary characteristic of a Class B building is the reinforced concrete frame in which the columns and beams can be either formed or precast concrete. They may be mechanically stressed. It is a fire-resistant structure.

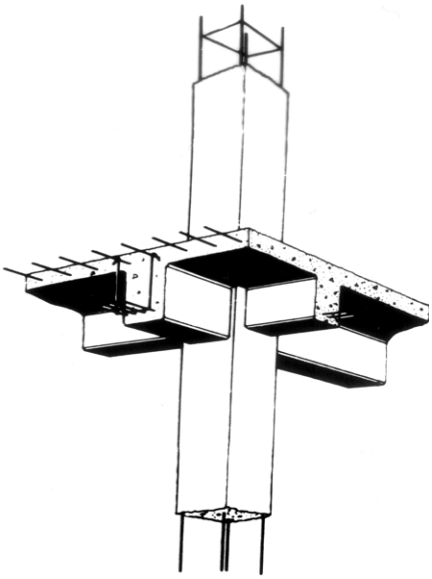
Floors and roofs in Class B structures are formed or precast concrete slabs. The exterior walls will generally be masonry or reinforced concrete curtain walls or any of the many types of wall panels of concrete, metal, glass or stone, etc. In some Class B buildings, the walls may be partially load-bearing. Interior partitions are often masonry, reinforced concrete or gypsum block, but many lightweight and movable partitions are used where structural walls are not needed.

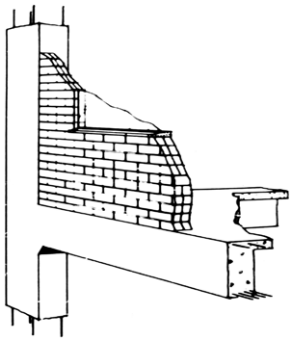
Included in this classification are Uniform, Basic and Standard Building Code Types I and II (noncombustible). This class is also referred to as Fire Resistive or Two – Four Hour Construction.

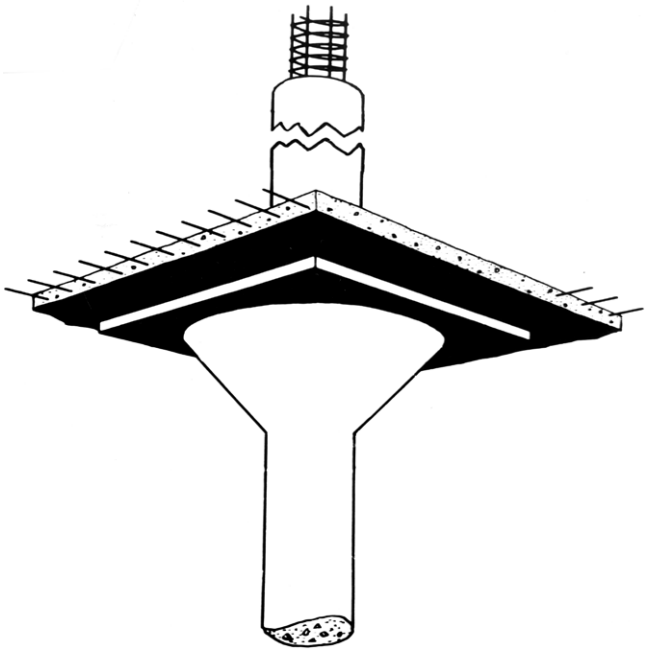












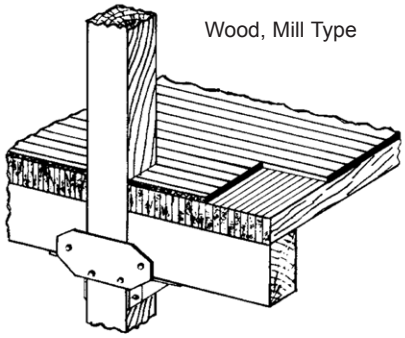
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CLASS C BUILDINGS

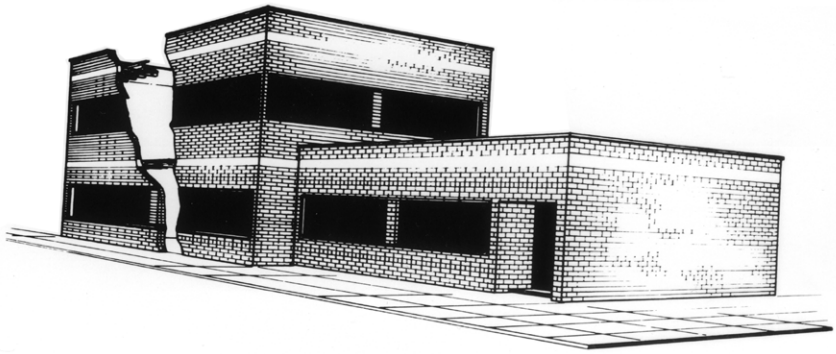
Class C buildings are characterized by masonry or reinforced concrete (including tilt-up) construction. The walls may be load-bearing, i.e., supporting roof and upper floor loads, or non-bearing with open concrete, steel, or wood columns, bents or arches supporting the load. Floors and roofs are supported on wood or steel joists or trusses, or the floor may be a concrete slab on the ground. Upper floors or roofs may be of concrete plank, steel deck, or wood.

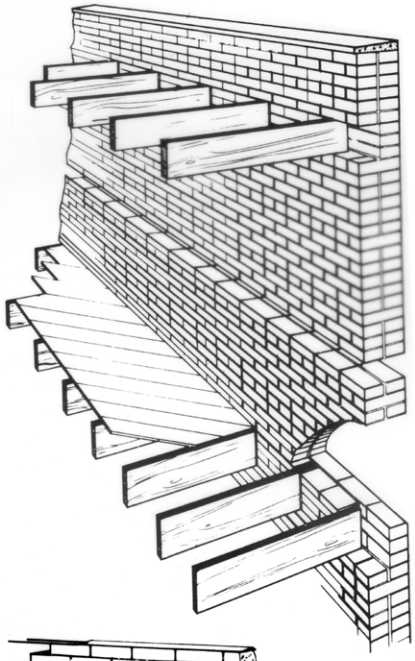
Bearing walls are frequently strengthened by concrete bond beams and pilasters. Included in this classification are Uniform and Basic building Code Type III (noncombustible wall) Standard Code Type V. This class is also referred to as Masonry or Unprotected Noncombustible, Joisted or Unprotected Masonry, or Ordinary or Unprotected One-hour and to include certain Two-Hour or Mill

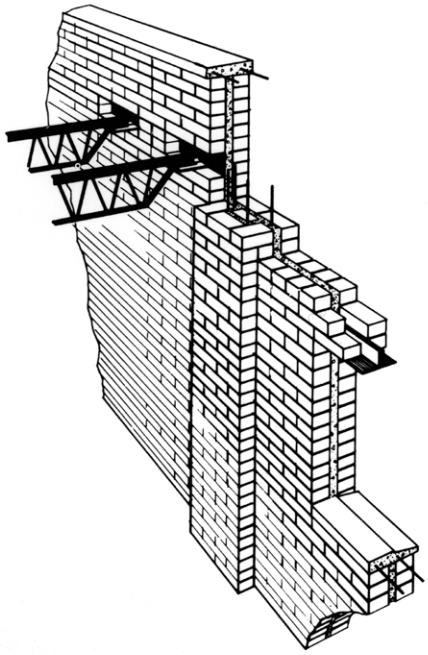
Construction (heavy timber). Class CMill type construction is characterized by thick masonry walls with heavy timber frames and laminated wood floors or concrete slab on the ground.

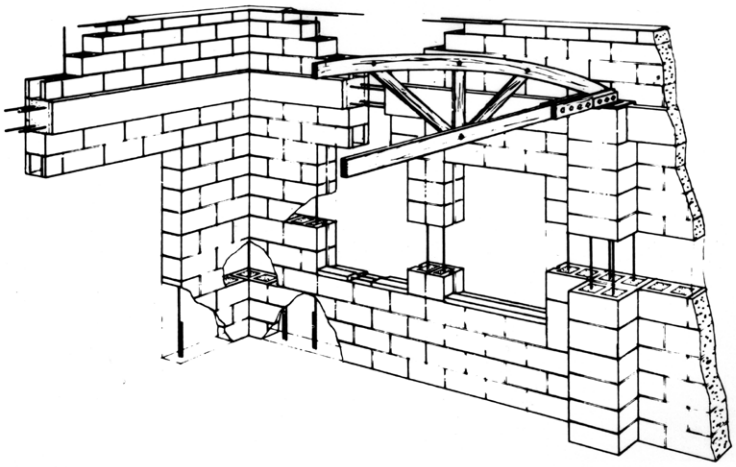


Wood, Mill Type





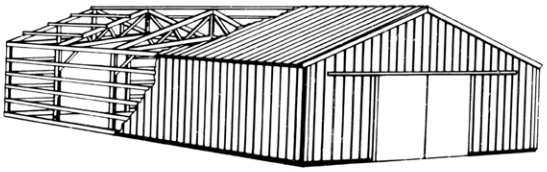




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CLASS D BUILDINGS

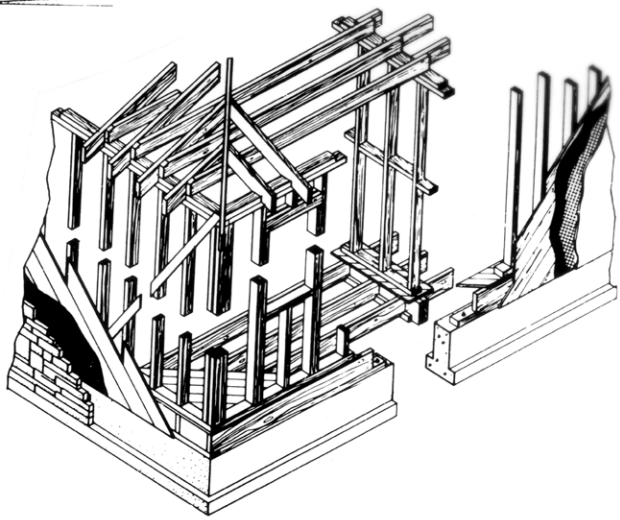
Class D buildings are characterized by combustible construction. The exterior walls may be made up of closely spaced wood or steel studs, as in the case of a typical frame house, with an exterior covering of wood siding, shingles, stucco, brick or stone veneer, or other materials. Otherwise, they may consist of an open-skeleton wood frame on which some form of curtain wall is applied including the pre-engineered pole- or post-frame buildings.

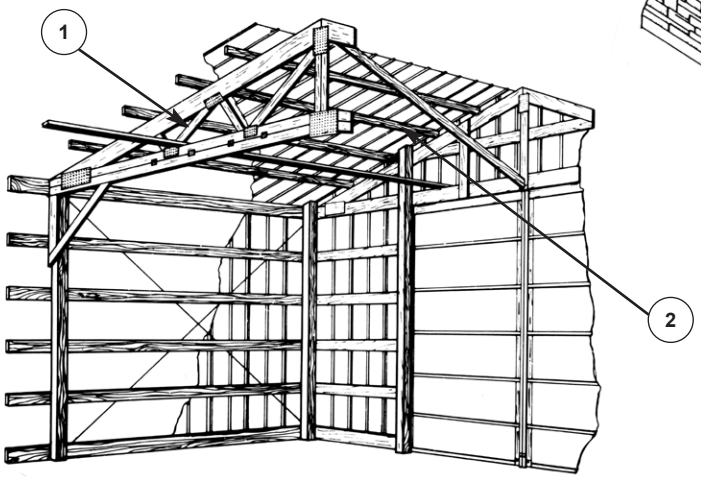


Floors or roofs are supported on wood or steel joists or trusses or the floor may be a concrete slab on the ground. Upper floors or roofs may consist of wood or metal deck, prefabricated panels or sheathing.

Construction Type V (wood-frame) of the Uniform, Type IV Basic and Type VI Standard Building Code are included in this classification. This class is also referred to as Unprotected-protected One Hour Construction.







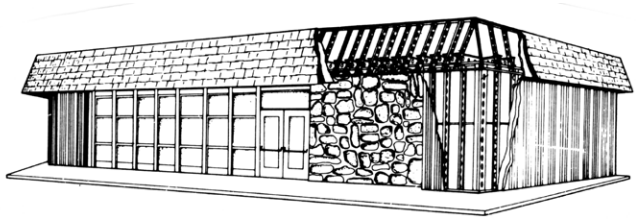
NOTES: **(1)** The cost of roof trusses such as these is included in the price of the frame costs for “wood, light pole-type construction”.

(2) The costs of purlins such as these can be found under the heading “light purlin supports only”. If these were heavier members, they would be costed as “open-wood system for corrugated metal”.

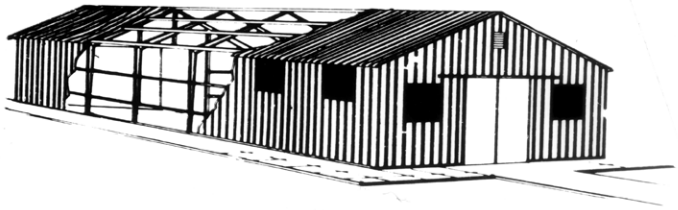
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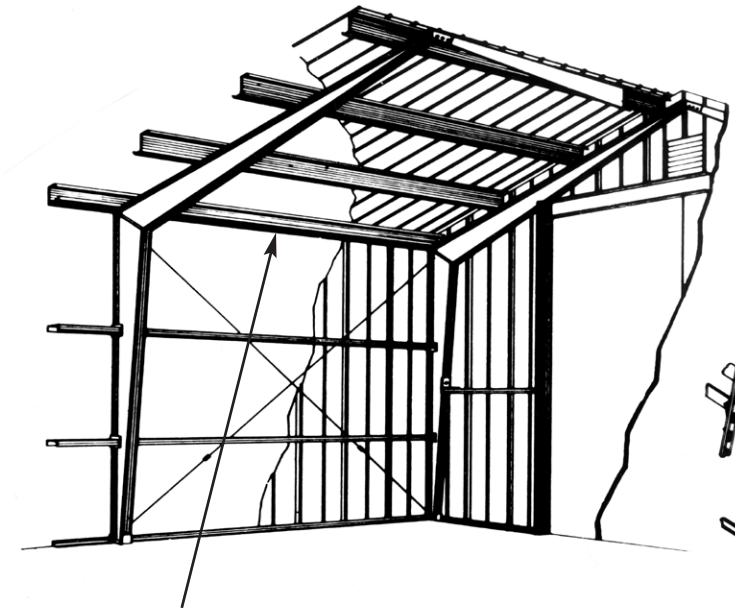
CLASS S BUILDINGS

Class S buildings are characterized by incombustible construction and prefabricated structural members. The exterior walls may be steel studs or an open-steel skeleton frame with exter or single or sandwich wall coverings consisting of prefabricated panels or sheet siding. Floors and roofs are supported on steel joists or beams, or the floor may be concrete slab on grade. Upperfloors or roofs may consist of metal deck, prefabricated panels or sheathing. Class S buildings can range from simple storage buildings to large “heavy” manufacturing buildings.

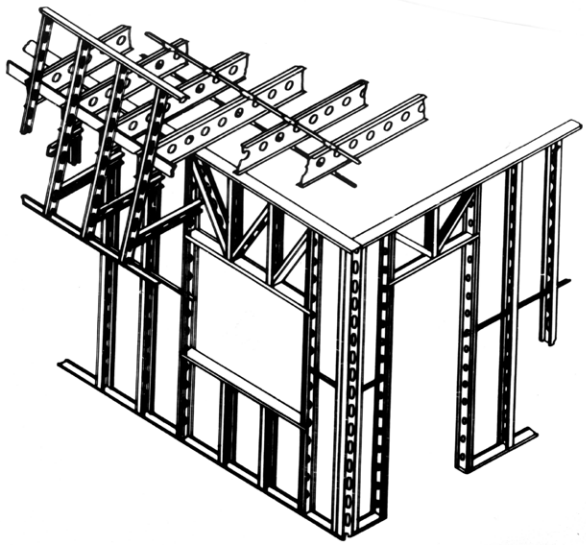


Included in this classific tion are Uniform and Standard Building Code construction, Type IV (noncombustible) Basic Code Type V. This class is also referred to as Noncombustible and can be One Hour Type II construction.





NOTE: These purlins are heavy enough to be priced as an “open-steel system for corrugated metal”.



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INTRODUCTION TO THE CALCULATOR METHOD

The Calculator Method is designed to be a fairly rapid meth od for obtaining square foot costs for typical buildings. Refinements are given for each occupancy, so that the base cost can be modified to fit deviations from the standard building described. If further refinements are needed, the various Segregated or Unit-in-Place sections may be used to adjust the costs.

The costs are classified by class and quality of construction. Buildings typical of a certain quality have many characteristics in common. For example, a Good quality building will usually have a good quality roof, so modifications for roof differences on a quality classified building are seldom necessary. The following are the most important square foot cost modifications. Many other modifications are possible, and since they are seldom cost important in relation to the additional time required to count and measure, they have been omitted from the Calculator Method. The base cost refinements that can be applied to buildings that vary from the general descriptions are as follows:

HEATING AND COOLING

Each heating and cooling cost is an average cost for the entire building described. To adjust for variations in the heating and cooling type found, take the difference between the type indicated and the type found in the subject building and add to or subtract from the base cost. If only a portion of the building is heated or cooled, then a prorated amount is to be used.

The costs for the heating and cooling systems listed for each ocupancy are based on the capacity, complexity and typical occupancy load, adjusted to Michigan's climate, and include a pro-rata share of contractors' overhead and profit and architects' fee.

ELEVATORS

The standard building description will indicate with an asterisk (in the Mechanical column) that an allowance was made for elevators. If the building under consideration has no elevators and the appropriate cost is marked with an asterisk, subtract the elevator cost (found on the refinement page) from the base square foot cost. Care must be exercised when using square foot costs where building sizes may fall outside a normal range of area served.

If elevators are found in a building not marked as typically having one, add the cost as a Lump Sum Adjustment from Section UIP 8.

Basement and mezzanine costs do not allow for elevators. Where elevator stops are found, add the cost per stop as a lump sum from the refinement pages, or see Section UIP 8.

SPRINKLERS

The basic building costs do not include sprinkler systems. If sprinkler systems are found, add from the corresponding refinement page. For more detail on sprinkler costs, refer to the appropriate SEG section.

HEIGHT

All base costs are for a typical base story height for that occupancy and for any basements associated with that occupancy. Any variation from this base height can be made by an adjustment found on the refinement page listed under Story Height Multipliers. For multistory buildings, an average story height can be used. The height adjustment is for wall height and not ceiling height.

SIZE AND SHAPE

The square foot cost is developed from variations in size and shape of a building. This is due to the variation in the proportion of exterior wall to total floor area. To adjust for this variation in cost, a Floor Area Perimeter table is provided, which gives a multiplier for various floor area and wall perimeter ratios. Most buildings being evaluated will not have the exact area and perimeter shown on the table, and some interpolation will be necessary. Usually the multiplier can be accurately approximated without going through a detailed interpolation. To enter the Floor Area/Perimeter table for multistory buildings, use the average floor area and the average perimeter.

MULTIPLE STORY BUILDINGS

The base costs given are for buildings of three stories or fewer. For buildings over three stories above ground, a recommended percen age adjustment is shown on the refinement page. This percent adjustment is based on the net of increased frame weight, construction difficulty, hi-rise wages, etc., less savings from shorter heating and plumbing runs, a single roof, etc. This added cost is applied to all floors, including basements, regardless of occupancy. In using the standard form, it is applied as a multiplier equal to one plus the percentage increase that is included on the refinement page.

BASEMENTS

Basements should be computed separately from the upper floors and are subject to their own modifiers and multipliers except in multistory buildings where they additionally receive the same multistory multipl er as the balance of the building. Costs are set to the typical base height for the occupancy and must be refined to the correct height. Some average cost parameters for typical basement types (i.e. utility, parking, display, etc.) are listed on the Calculator Pages under specific occupancies where commonly encountered. These costs can be used for all occupancies within the group, where appropriate.

Finished basements, i.e., those containing apartments, retail stores, etc., as a general rule of thumb will cost approximately 75% to 80% of the comparable aboveground portion of the building. Semibasements that are half exposed will cost 85% to 90% of the same figure.

MEZZANINES

Mezzanine floors are computed separately and, since they do not include any exterior wall or heat, are not subject to any modifiers or multipliers for size and shape or height. Some average cost parameters for typical mezzanine types (i.e., storage office, display, etc.) are listed on the Calculator Pages under specific occupancies where commonly encountered. These costs can be used for all occupancies within the group, where appropriate.

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COMMENTS AND EXPLANATIONS

The costs on the Calculator pages are averages of detailed estimates, actual breakdowns, and total end costs of many actual construction projects. These costs are assembled into groups by typical occupancy and general quality and each is adjusted to fit the base description. The only items adjusted are those outlined above. All other construction components are considered as co mensurate with the general quality of the building. There are a number of construction components that affect the total cost of a building and to take them all into consideration would entail a complete, detailed estimate. The above refinements are provided as the ones that have significant effect on the total building cost, and can be computed readily, thus providing an accurate estimate in a reasonably short time. Those wishing to give more detailed consideration to additional construction components may use the Segregated Cost Method, Sections SEG 1 through 6, or for further refinement, use the various Unit-in-Place costs found in Sections UIP 1 through 8.

County Multiplier For UIP Sections

COUNTY MULTIPLIER FOR UIP SECTIONS

The county multiplier to be applied to UIP costs should be selected based on the following:

1. If the UIP cost is for a component of a building being priced, use the same county multiplier for the UIP costs as is being used for the building.

2. If the UIP cost is not associated with a building being priced use the following:

Nature of Item Being Priced	Multiplier to Use
Structural steel (fireproofed)	Class A
Reinforced concrete	Class B
Masonry	Class C
Wood on wood or steel stud	Class D
Prefabricated steel	Class S

3. If neither #1 nor #2 works, the assessor must use his/ her judgement as to which multiplier to use or whether an average multiplier is appropriate.

CALCULATOR EXAMPLE

A filled-in field form and sample pages from which the prices in the examples are taken are shown in Figures 1 through 4. The subject building used in the example is a 3-story, Goodquality, Class C apartment building with brick exterior and no elevators or sprinklers. This building is priced from the page labeled Apartments. It is of heavier construction than the type found under Multiple Residences. The dimensions are 50’ by 100’, with a height of 33’. This information, along with the age and condition of the building, is entered on Lines 4 through 11 of the form. The area and perimeter are computed on the back of the form as shown in Figure 2.

The base square foot cost, $174.00, is entered on Line 12 from the Calculator Cost page (Figure 3). In this example, the subject building has a heat pump system. An amount for elevators is included in the base costs, and since the subject building has no elevators, a deduction of $5.47 must be made (Figure 4) and entered on Line 14. There were no miscellaneous items to add, so that the adjusted square foot cost is $174.00 minus $5.47, or $168.53. This is entered on Line 16. The number of stories (three) is the base figure, so the multiplier on Line 17 is 1.00. The 11-foot average story height is more than the 10-foot base height, so the correct multiplier, 1.027, is taken from the refinement page (Figure 4) and entered on Line 18. The subject has an average floor area of 5,000 square feet, and an average perimeter of 300 feet. Since 5,000 square feet is not on the table, but is between 4,000 square feet and 6,000 square feet, a multiplier can be found for it.

Entering the table at a 300-foot perimeter, the multipliers are 1.021 and .959 respectively, and the interpolated value is .990, which is then entered on Line 19. Line 18 is multiplied by Line 19 and the answer is placed on Line 20. Line 16 is then multiplied by Line 20 and the answer of $171.40 (the refined square foot cost) is entered on Line 21.

The county multiplier is assumed to be .980 for this example, and is entered on Line 22. Line 21 is multiplied by Line 22 to give the final square foot cost on Line 23, and this is multiplied by the total square footage of floor area (calculated on the back of the form).

The answer, $2,534,550 is placed on Line 25. In this example, there were no Lump Sum Additions, so Line 26 is left blank, and the sum of Lines 25 and 26 is entered on Line 27 as the reproduction cost.

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Michigan Department of Treasury
L-4104 (Rev. 10-91)

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STC CALCULATOR COST COMPUTATION SHEET (S.F. COSTS)

1.	OWNER	TENANT				PROPERTY CODE
2.	TYPE	YEAR BUILT	YEAR REMOD.	% GOOD	CAL PAGE	PROPERTY ADDRESS
3.	COUNTY	UNIT			EXAMINED BY	DATE

	SECTION I	SECTION II	SECTION III	SECTION IV
4. Occupancy	Apartment			
5. Building class and quality	Cls. C / Qual. 3	Cls. / Qual.	Cls. / Qual.	Cls. / Qual.
6. Exterior wall				
7. No. of stories & height per story	No. 3 / Ht. 11	No. / Ht.	No. / Ht.	No. / Ht.
8. Average floor area	5000			
9. Average perimeter	300			
10. Year built, year remodeled	Blt. / Remod.	Blt. / Remod.	Blt. / Remod.	Blt. / Remod.
11. Percent condition	%	%	%	%

	SECTION I	SECTION II	SECTION III	SECTION IV
12. **Base Square Foot Cost**	183.50			
SQUARE FOOT REFINEMENTS				
13. Heating, cooling, ventilation (Base)	-----			
14. Elevator deduction (Use Lump Sum)	5.47			
15. Miscellaneous				
16. Total lines 12 through 15	188.97			
HEIGHT AND SIZE REFINEMENTS				
17. Number of stories-multiplier (3 Story Building)	1,000			
18. Height per story-multiplier (see line 7)	1.027			
19. Floor area-perimeter-multiplier (see lines 8 and 9)	.990			
20. Combined height and size multiplier (lines 17 x 18 x 19)	1.017			

FINAL CALCULATIONS	SECTION I	SECTION II	SECTION III	SECTION IV
21. Refined square foot cost (line 16 x 20)	192.18			
22. County multiplier	.980			
23. Final sq. ft. cost (line 21 x line 22)	188.34			
24. Area (Back of this form)	15,000			
25. Line 23 x line 24	$2,825,100			
26. Lump sums (line 35, back of sheet)				
27. Reproduction/Replacement Cost (line 25 + line 26)	$2,825,100			
28. Physical Depreciation multiplier (Assessors Manual)				
29. Functional obsolescence multiplier (Percent Good)				
30. Economic obsolescence multiplier (Percent Good)				
31. Depreciated Cost (Multiply line 27 x 28 x 29 x 30)				
32. Economic Condition Factor (E.C.F.)				
33. True Cash Value				

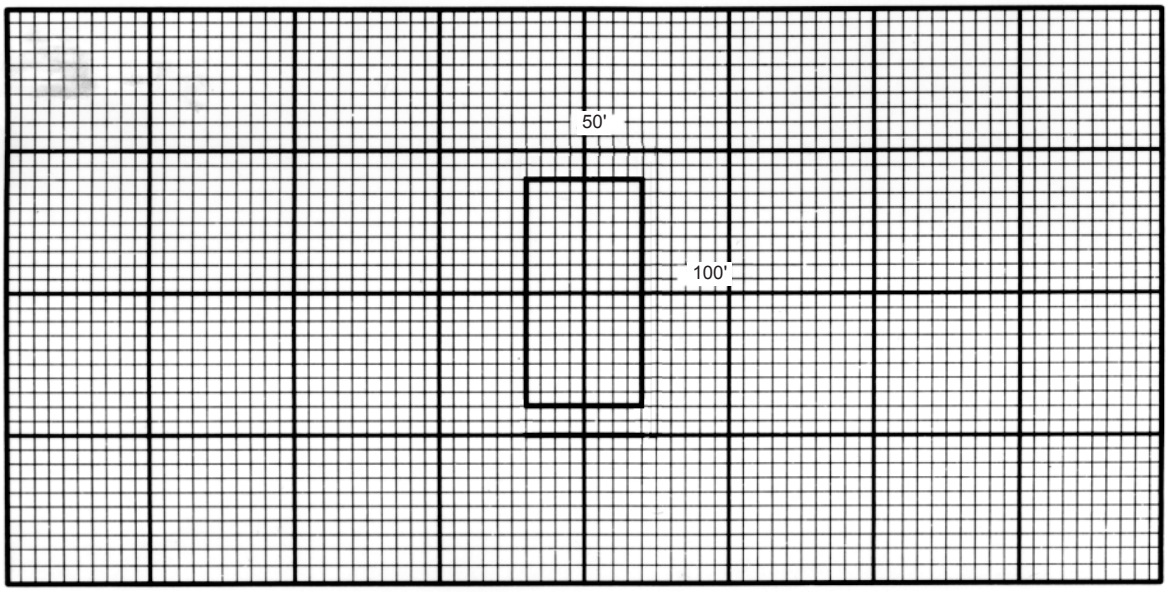
TOTAL OF ALL SECTIONS

34. Reproduction/Replacement cost $2,825,100 Depreciated cost _____ True Cash Value _____

See back of form for drawings and area and lump sum value calculations.

Figure 1

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Calculations (Area, etc.) 50' x 100' = 5,000 sq. ft. x 3 floors = 15,000 sq. ft.

35. **LUMP SUMS (Sprinklers, Elevators, etc.)**

Section	Item	Cost	Section	County	Architect Fees	Composite	Area	Total Cost
Section I								
Section II								
Section III								
Section IV								
	TOTAL							

Figure 2

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CAL 2

APARTMENTS



AVERAGE CLASS A



AVERAGE CLASS B

OCCUPANCY DESCRIPTION: High-rise apartments are structures with three or more stories of multiple dwelling units. Each dwelling unit consists of its own separate living area and kitchen facility.

INCLUDED IN COSTS: Architects' fees and contractors' overhead and profit. Allowance for lo by area and interior hall access to dwelling units. Elevators included where designated with an (*) asterisk

NOT INCLUDED IN COSTS: Sprinklers, appliances or balconies.

SQUARE FOOT COST TABLE

CLASS	TYPE	COST/ SQ. FT.	EXTERIOR WALLS	INTERIOR FINISH	LIGHTING, PLUMBING AND MECHANICAL	HEAT
A	Good	$ 255.00	Face brick, limestone, metal or concrete and glass panels	Good interior detail, carpet, or hard-wood, sheet vinyl or ceramic tile	*Many good fixtures, TV jacks good baths and kitchens	Warm and cool air (zoned)
	Average	203.50	Little trim, brick, block, metal or concrete and glass	Drywall or plaster, average carpet and vinyl composition floors	*Few electric fixtures, average plumbing, one bath per unit	Heat pump system
	Low Cost	161.50	Very plain, brick or block, or low-cost concrete panels	Drywall, sprayed or painted ceilings, very plain, asphalt tile	*Minimum uniform code, one bath per unit	Hot water
B	Good	228.00	Face brick, limestone, metal or concrete and glass panels	Good interior detail, carpet or hard-wood, sheet vinyl or ceramic tile	*Many good fixtures, TV jacks, good baths and kitchens	Warm and cool air (zoned)
	Average	181.50	Little trim, brick, block, metal or concrete and glass	Drywall or plaster, vinyl composition tile and average carpet	*Few electric fixtures, average plumbing, one bath per unit	Heat pump system
	Low Cost	144.50	Very plain, brick or block or low-cost concrete panels	Drywall, sprayed or painted ceilings, very plain, asphalt tile	*Minimum uniform code, one bath per unit	Hot water
A-B	Basement units	160.00	Half exposed, good fenestration	Drywall or plaster, painted, vinyl composition and average carpet	Few electric fixtures, average plumbing, one bath per unit	Hot water
	Finished Basement	103.50	Finished interior	Finished floor and ceilings, game room	Adequate lighting/plumbing	Forced air
	Parking Basements	81.50	Unfinished interior	Concrete w/ hardener, lines & stops	Minimum lighting, floor drains	Ventilation
	Utility Basement	71.00	Painted interior	Utility and storage areas	Utility lighting and plumbing	None
C	Good	183.50	Brick, metal or concrete and glass panels, some trim	Drywall or plaster, good carpet or hardwood, vinyl composition, cer. tile	*Good electric/plumbing, good baths and kitchens	Heat pump system
	Average	138.50	Little ornamentation, brick or block, concrete panels	Drywall or plaster, average carpet and vinyl composition	*Average fixtures, one bath per unit, average circuits	Forced air
	Low Cost	104.00	Very plain brick or block, minimum fenestration	Drywall and painted masonry/plaster coat, asph. tile, low-cost carpet	Minimum uniform code, one bath per unit	Indiv. thru-wall heat pumps
CMILL	Good	207.00	Mill type construction, heavy brick wall, trusses, good sash	Drywall, or plaster, good carpet or hardwood, vinyl comp., ceramic tile	*Good electric/plumbing, good baths and kitchens	Heat pump system
	Average	168.50	Mill type construction, brick and block, wood trusses	Drywall, or plaster, average carpet and vinyl composition, softwood	*Average fixtures, one bath per unit, average circuits	Hot water
D	Good	174.00	Good stucco or siding with ornamentation	Good drywall or plaster, carpet, good vinyl composition tile or sheet	*Good fixtures, many outlets, over one bath per unit	Heat pump system
	Average	131.50	Frame and stucco, little trim, standard design	Plaster or drywall, carpet or hardwood, vinyl composition tile	*Average fixtures, one bath per unit, average circuits	Forced air
	Low Cost	98.00	Low-cost siding or stucco	Drywall, low-cost carpet, asphalt tile	Minimum uniform code, one bath per unit	Indiv. thru-wall heat pumps
D MASONRY VENEER	Good	179.00	Good brick veneer with ornamentation	Good drywall or plaster, carpet, good vinyl composition tile or sheet	*Good fixtures, many outlets, over one bath per unit	Heat pump system
	Average	135.50	Brick veneer, little trim, standard design	Plaster or drywall, carpet or hardwood, vinyl composition tile	*Average fixtures, one bath per unit, average circuits	Forced air
S	Good	169.50	Best insulated sandwich walls, good fenestration, good frame	Drywall, carpet and vinyl composition, ceramic tile	*Good fixtures, many outlets, over one bath per unit	Heat pump system
	Average	127.00	Sandwich walls, fenestration to code, little trim	Drywall, carpet and vinyl comp., average cabinetry and finish	*Average fixtures, one bath per unit, average circuits	Forced air
CDS†	Basement units	115.50	Half exposed, good fenestration	Drywall or plaster, vinyl composition tile and average carpet	Few electric fixtures, average plumbing, one bath per unit	Forced air
	Finished Basement	63.50	Finished interior, add for saunas, pools, bars and equip.	Gypsum board ceiling, vinyl composition tile sheet vinyl	Adequate lighting/plumbing	Electric wall heaters
	Parking Basement	51.50	Unfinished interior	Plaster or drywall ceiling, concrete floor	Minimum lighting, floor drains	Ventilation
	Utility Basement	44.25	Unfinished interior	Unfinished floor and ceiling	Minimum lighting/plumbing	None

†For fire-resistant Type I basements, with concrete slab separation under Class C, D or S units, add $8.79 per square foot to the basement cost.

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Figure 3

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APARTMENTS

REFINEMENTS: On this page are adjustments to the base costs from the previous page. Follow steps 1 through 5 to obtain final costs.

1

BUILT-IN APPLIANCES: For individual listing, see Segregated Cost, Section SEG 1.

	LOW	AVG.	GOOD	EXCL.
Allowance (if not itemized)	1820.00	4150.00	7550.00	12800.00

ELEVATORS: Buildings whose base costs include elevators are marked with an asterisk (*). If the building under consideration has no elevators, deduct the following from the base costs. For detailed costs see Section UIP 8.

Classes A/B	Sq. Ft. Costs	Class C	Sq. Ft. Costs	Classes D/S	Sq. Ft. Costs
Good	8.39	Good	5.47	Good	4.86
Average	6.43	Average	4.42	Average	4.06
Low Cost	4.86				

ELEVATOR STOPS: For basement stops, add $ 9,550 to $14,400 per stop.

SPRINKLERS: Apply to area covered by sprinklers.

Sq. Ft.	LOW	AVG.	GOOD	EXCL.
3,000	$ 5.14	$ 6.63	$ 8.55	$ 11.05
5,000	4.75	6.09	7.80	10.00
10,000	4.28	5.44	6.91	8.77
20,000	3.87	4.87	6.10	7.69
50,000	3.36	4.20	5.24	6.50
100,000	3.00	3.73	4.59	5.71
200,000	2.69	3.30	4.06	5.01

BALCONIES: Balcony costs include the supporting structure, decking and rails. Apply costs to the sq.ft. of the balcony area.

	LOW	AVG.	GOOD	EXCL.
Concrete	$ 34.00	$ 43.50	$ 55.00	$ 71.00
Steel	31.75	42.50	56.50	75.50
Wood	29.75	39.50	52.50	68.50
Add for ornate finishes, balustrades	29.50	36.00	44.25	54.00
Add for roofs or awnings	15.35	20.40	27.00	36.00

2

HEATING AND COOLING

These costs are averages of the total installed cost of the entire heating or cooling system including a prorated share of contractors' overhead and profit and architects' fees. If the heating found in the building being assessed is different from that indicated, take the difference between the costs of the two and add to or subtract from the base square foot cost. For other types or system adjustments, see Segregated Costs.

HEATING ONLY	Sq. Ft. Costs	HEATING & COOLING	Sq. Ft. Costs	COOLING ONLY	Sq. Ft. Costs
Electric cable or baseboard	$ 10.50	Package A.C. (short ductwork)	$ 21.90	Central refrigeration (zoned)	$ 16.00
Electric wall heaters	3.84	Warm and cool air (zoned)	30.50	package (short ductwork)	10.65
Forced air furnace	15.15	Hot/chilled water (zoned)	45.50	Central evaporative	7.04
Hot water, baseboard/convector	20.75	Heat pump system	25.50	Pkg. refrig.	$2,750 to $3,425 per ton capacity
radiant floor/ceiling	22.55	Ind. thru-wall heat pumps	10.75	Evap. coolers.	$331 to $535 per MCFM capacity
Steam (including boiler)	19.20				
without boiler	16.85	Small indiv. heat pumps cost $2,280 to $3,125		**VENTILATION ONLY**	
Wall or floor furnace	4.43	per ton of rated capacity.		Vent. (blowers/ducts)	$ 3.18

3

HEIGHT REFINEMENTS

MULTISTORY BUILDINGS: Add .5% (1/2%) for each story over three, above ground, to all base costs.

STORY HEIGHT MULTIPLIERS: Multiply base cost by the following multipliers for any variation from the average story height.

Average Wall Height	Square Foot Multiplier	Average Wall Height	Square Foot Multiplier
8	.947	12	1.055
9	.973	13	1.064
10	1.000 (base)	14	1.114
11	1.027		

4

Average Floor Area Sq. Ft./Story	AVERAGE PERIMETER 200	250	300	350	400	450	500	550	600	650	700	750	800	850	Average Floor Area Sq. Ft./Story
2,000	1.083	1.143	1.206	1.267	1.329	-----	-----	-----	-----	-----	-----	-----	-----	-----	2,000
4,000	.959	.989	1.021	1.052	1.083	1.113	1.143	1.175	1.206	1.237	1.268	1.299	1.329	-----	4,000
6,000	-----	.938	.959	.980	1.000	1.021	1.042	1.163	1.083	1.103	1.123	1.144	1.165	-----	6,000
8,000	-----	-----	.929	.944	.959	.974	.989	1.005	1.021	1.037	1.052	1.068	1.083	-----	8,000
10,000	-----	-----	-----	.922	.935	.947	.959	.972	.984	.997	1.009	1.022	1.034	-----	10,000
12,000	-----	-----	-----	-----	.917	.928	.938	.949	.959	.970	.960	.990	1.000	-----	12,000
14,000	-----	-----	-----	-----	-----	.915	.924	.933	.942	.951	.959	.968	.977	.986	14,000
16,000	-----	-----	-----	-----	-----	.906	.913	.921	.929	.937	.944	.952	.959	.967	16,000
18,000	-----	-----	-----	-----	-----	.897	.904	.911	.917	.925	.932	.939	.945	.952	18,000
20,000	-----	-----	-----	-----	-----	-----	.898	.904	.910	.916	.922	.929	.935	.941	20,000
22,000	-----	-----	-----	-----	-----	-----	.892	.898	.903	.909	.915	.920	.926	.932	22,000
25,000	-----	-----	-----	-----	-----	-----	.883	.888	.892	.897	.903	.907	.912	.916	25,000

5 **USE COUNTY MULTIPLIERS IN MULTIPLIER SECTION.**

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Figure 4

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